

Pruthvi Prodduturi

Engineering Lead

LinkedIn: www.linkedin.com/in/pruthviprodduturi

Pruthvi.Prodduturi@gmail.com +1 330-785-5526 | Redmond, WA

PROFESSIONAL SUMMARY

Senior software engineer with 11+ years across distributed data systems, query engines, and zero-to-one platform engineering. Authored a distributed SQL engine (MPP) performance analysis that was escalated from the Azure Data President to CEO Satya Nadella and influenced the engine roadmap. Built the governed lakehouse behind Microsoft's AI competitive intelligence powering the M365 Copilot analytics plane — billions of rows/month, 600+ teams — and designed a real-time IoT orchestration platform from zero for autonomous datacenter robots, with environmental monitoring over 18 PB/month of datacenter telemetry (~630B rows/day). Fluent from query-engine internals to cloud-native service architecture, and effective communicating technical direction to executive audiences.

CORE SKILLS

Distributed Systems: MPP query engines, columnar storage, vectorized execution, partitioning strategies, async job systems, IoT orchestration

Data Architecture: Lakehouse design, semantic models, telemetry pipelines, ML scoring, stream/batch processing, data governance, exfiltration prevention, EUDB compliance

Data Engineering: SQL optimization, Presto, Trino, StarRocks, ClickHouse, Fabric SQL, ADF, Synapse, ADLS, Cosmos DB

Platform Engineering: Python, FastAPI, React, Azure (App Service, IoT Hub, Blob Storage), managed identity, VNet

Leadership: Zero-to-one platform builds, cross-geo leadership, mentoring engineers, executive technical communication, operating under ambiguity

EXPERIENCE

Microsoft

Redmond, WA

Senior Software Engineer

Nov 2022 – Present

Distributed Systems & Query Engine Analysis

- **Authored** a deep performance analysis of Microsoft's distributed SQL engine (MPP architecture, columnar storage) covering query optimization, predicate pushdown, partition pruning, and materialization strategies. Escalated from the Azure Data President to CEO Satya Nadella. Directly influenced the engine roadmap.
- **Hands-on production experience** across distributed query engines — Fabric SQL, Presto, Trino, StarRocks, ClickHouse, Apache Kylin — with a working understanding of the tradeoffs between MPP, vectorized execution, columnar storage, and lakehouse query patterns from building on all of them.

Data Platform Architecture (Billions of Rows / Month)

- **Built** the lakehouse architecture (ingestion → transformation → semantic model → serving) powering Microsoft's AI competitive intelligence powering the M365 Copilot analytics plane — replaced 5+ fragmented org-specific pipelines with a single governed platform serving 600+ teams.
- **Architected** pipelines processing billions of rows/month across 10+ GenAI products (Copilot, ChatGPT, Gemini, Claude, Perplexity), standardizing desktop, mobile, SSO, and connector signals into governed, query-ready models.
- **Established** a unified metrics framework — KPI computation, LLM evaluation workflows, and ML-driven scoring consumed in weekly executive scorecards driving Microsoft's AI competitive strategy.
- **Delivered** sub-5-second query latency on competitive dashboards by designing a semantic model that eliminated scan-heavy patterns — replacing a system of ad-hoc queries and spreadsheets with governed, reproducible analytics.
- **Reduced** SLA breaches by 25% through pipeline reliability engineering and data governance hardening.

Platform Engineering — IoT Fleet Orchestration (2026)

- **Designed and built** a real-time orchestration platform from zero for autonomous datacenter robots — 26-endpoint REST API (Python/FastAPI), React command center, per-device job queue, and mission lifecycle engine controlling physical hardware via IoT Hub direct methods.
- **Instrumented** real-time environmental monitoring over 18 PB/month of datacenter telemetry (~630 billion rows/day) across distributed datacenter sites.
- **Defined** cross-team API contracts between the cloud platform and robotics firmware team, replacing a CLI-as-API approach with proper service boundaries — working through undocumented firmware behavior, schema changes, and hardware version fragmentation.
- **Engineered** parallel fleet discovery (asyncio + thread pooling), reducing $O(n)$ sequential device calls to a single parallel call and architected to scale to 1000+ devices. Built auth with token deduplication — 8 concurrent requests resolved to one identity call via async futures.
- **Hardened** the production architecture to zero-key operation — managed identity for all service-to-service auth, private endpoints for storage, and VNet integration for network isolation.
- Architect of the IDEAS–Fabric Partnership
- The sole engineer behind Microsoft’s IDEAS journey onto Fabric — from the initial Trino-vs-Fabric evaluation that catalyzed the partnership to the semantic model strategy powering analytics at scale. Designed and implemented the Direct Lake semantic model approach, the Fabric development lifecycle for compliance and reliability (Git-integrated CI/CD, semantic workspace isolation, production promotion via Azure DevOps). Architecture published on Microsoft Learn as the reference for Fabric adoption at enterprise scale.
- Data Security & Compliance at Platform Scale
- Surfaced a critical data exfiltration vulnerability in Microsoft Fabric, authored the end-to-end remediation — access-boundary design, asset-naming standards, and governance policies — and presented the required API changes and requirements to the platform team. Design adopted as a native Fabric capability. Laid foundational standards for EUDB-compliant data handling, defining how data residency and sovereignty controls are enforced at platform scale.

Software Engineer II

Jan 2019 – Nov 2022

- **Delivered** business metrics, data pipelines, and distributed reporting systems for Microsoft News.
- **Built** durable OLAP solutions using Apache Kylin, Mondrian, and ClickHouse for high-volume analytics.
- **Partnered** with COSINE to build a unified hot-data platform using Azure Kubernetes Service and Presto, improving performance and stability.
- **Accelerated** data availability by 7+ hours/day through optimized reporting pipelines.
- **Developed and maintained** CI/CD pipelines, improving deployment consistency and reducing operational overhead.

EARLIER EXPERIENCE

Technical Lead — HCL America

2017 – 2019

Reduced ETL processing time by 95% using C#, Avocado, and SCOPE automation; led offshore telemetry and analytics teams supporting MSN.

Senior Software Engineer — Mindtree

2013 – 2016

Delivered BI analytics and telemetry systems for Microsoft IDC; built SCOPE-based ETL pipelines reducing reporting time by 85%; migrated Power Pivot to Power BI, reducing analysis time by 80%.

EDUCATION

Kent State University — M.S., Digital Sciences (Computer & Information Sciences) 2016 – 2017

Jawaharlal Nehru Technological University — B.Tech., Electronics & Communication Engineering